ex 18

English Recent Login Sign up

```
% Hexagonal Cell Structure & Capacity Planning Simulation
%
% AIM:
% 1. Simulate hexagonal cellular structure
% 2. Calculate coverage area, cell radius, and user capacity
% 3. Visualize frequency reuse patterns and user distribution
% 4. Analyze effect of frequency reuse (K) on network capacity

clear;
close all;
clc;

%% Simulation Parameters

cellRadius = 500;              % Radius in meters
numTiers = 1;                  % 1-tier hex = 7 cells
totalUsers = 300;              % Total users
bandwidth = 10e6;              % 10 MHz bandwidth
spectralEfficiency = 2;        % 2 bps/Hz

clusterSize_K = 3;             % Frequency reuse factor

% Reuse colors
reuseColors = ['r','g','b','m','c','y'];

%% Derived Parameters

coverageArea = (3*sqrt(3)/2) * cellRadius^2;

capacity_per_cell = ...
    (bandwidth / clusterSize_K) * spectralEfficiency;

fprintf('\n----- CELL PARAMETERS -----\n');

fprintf('Cell Radius: %.2f m\n', cellRadius);

fprintf('Coverage Area: %.2f sq.m\n', coverageArea);
```

Run Save Output Input

```
----- USER DISTRIBUTION -----
Cell 1: 81 users
Cell 2: 29 users
Cell 3: 39 users
Cell 4: 30 users
Cell 5: 42 users
Cell 6: 36 users
Cell 7: 43 users
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----- CAPACITY VS REUSE FACTOR -----
K = 1 --> Capacity = 20.00 Mbps
K = 3 --> Capacity = 6.67 Mbps
K = 4 --> Capacity = 5.00 Mbps
K = 7 --> Capacity = 2.86 Mbps
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Simulation completed successfully.

[Execution complete with exit code 0]
```